

Assessment of the Attached Weak Simplex Conjecture Proof

Executive summary

The attached manuscript proves the **weak simplex conjecture in Cover's equal-energy formulation**: for $m = n + 1$ unit vectors $x_1, \dots, x_m \in S^{n-1}$, the regular simplex maximizes the average maximum-likelihood decoding success probability in additive Gaussian noise, for every signal strength $\lambda > 0$. The manuscript does this by rewriting decoding success as an exponential moment of a Gaussian maximum, normalizing the regular simplex to independence, proving a **centered log-concave Gaussian product inequality**, and then using an **inverse-Mills-ratio variational tilt** to convert that product inequality into a Gaussian orthant comparison and hence stochastic domination of maxima. [filecite turn0file0](#)

My bottom-line assessment is:

The proof is not novel as the first claimed proof of the weak simplex conjecture. An earlier full proof claim already exists: Adriano Pastore's 2023 arXiv paper, which explicitly claims to solve the weak simplex conjecture by a different relaxation-and-symmetry method. In addition, several partial or related results were already known: Kabluchko-Litvak-Zaporozhets gave the standard Gaussian/mean-width reformulation in 2015, Sun-Hu-Lan solved the $m = 4$ Gaussian-maximum case in 2020, and Goldsmith posted a 2021 all-dimensional proof claim for the **simplex mean-width conjecture**, which is weaker than the weak simplex conjecture. [1](#)

The manuscript does appear to contain a genuinely different proof architecture. Its central route through a product inequality for centered one-dimensional log-concave factors and an adaptive variational tilt is materially different from Pastore's Voronoi/Gaussian-centroid relaxation proof. On proof strategy alone, this looks substantially new. [filecite turn0file0](#) [2](#)

However, several key steps are already known or have very close prior analogues. The maximum-likelihood integral identity is standard in the modern literature on the weak simplex conjecture; the symmetric rectangle inequality in Section 5 is classical Šidák-Khatri territory; and the manuscript's Section 6 theorem is very close in flavor to the 2025–2026 centered Gaussian correlation / centered log-concave / forward-reverse Brascamp-Lieb developments of Nakamura-Tsuji and Milman-Nakamura-Tsuji, which explicitly emphasize centered log-concave data and self-convolution/CLT technology. [3](#)

I do not think the manuscript merely reduces the weak simplex conjecture to an already solved external problem. It does not simply reduce to the simplex mean-width conjecture, nor does it obviously collapse to the 2017 Balitskiy-Karasev-Tsigler simplex-measure reduction. Instead, it introduces its own internal reduction to a centered log-concave Gaussian product inequality plus an adaptive tilt lemma. That said, the Section 6 product inequality now lives very close to already-solved 2025–2026 centered Gaussian functional inequalities, so this part may no longer be truly novel in 2026 even if the manuscript proves it independently. [4](#)

I did not find an obvious fatal error on a first-pass mathematical audit, and the manuscript is more detailed than many proof claims in this area. But the part that most needs expert verification is Section 6, because that is both the paper’s main engine and the place of strongest overlap with recent centered log-concave Gaussian inequality machinery. The next most important check is whether Section 8’s variational construction can be recognized as an instance of an existing Gaussian-saturation framework.

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What the manuscript proves and how it is positioned

The manuscript’s exact statement is stronger than the usual mean-width formulation. It proves optimality of the regular simplex for **every** signal strength $\lambda > 0$, which after the standard maximum-likelihood integral identity becomes a comparison of exponential moments of Gaussian maxima rather than only first moments. In particular, the manuscript obtains a first-order stochastic domination statement for a normalized Gaussian maximum, not just the expectation comparison that underlies the simplex mean-width conjecture. filecite turn0file0 6

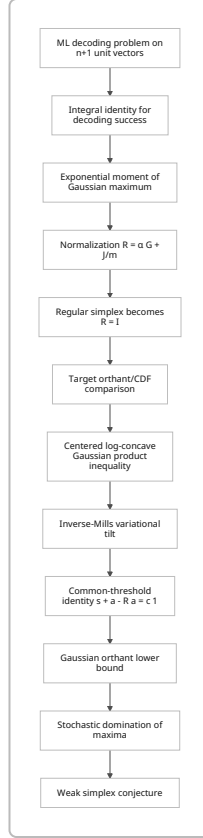
That matters historically. Kabluchko–Litvak–Zaporozhets explain that the simplex mean-width conjecture is equivalent to maximizing $\mathbb{E} \max_i \xi_i$ over centered Gaussian vectors with equal variances, with the regular simplex corresponding to the equiangular covariance $-1/(n - 1)$ off the diagonal. Goldsmith’s 2021 manuscript also explicitly notes that the weak simplex conjecture implies the simplex mean-width conjecture, not conversely. So even if one accepts all known mean-width proof claims, the attached manuscript still targets a stronger statement. 7

The broader historical context is also standard by now. The **strong** simplex conjecture was disproved by Steiner in 1994, while the **weak** simplex conjecture remained open and continued to attract proof attempts and reformulations. Balitskiy–Karasev–Tsigler’s 2017 survey paper states the modern equal-energy formulation and proposes reductions to Gaussian-measure problems; Pastore’s 2023 paper explicitly claims a full proof of the weak simplex conjecture by a different route. 8

The attached manuscript’s route can be summarized as follows. It turns decoding success into an exponential moment of a Gaussian maximum, transforms the regular simplex into independence by adjoining a common Gaussian coordinate, proves a centered log-concave factor inequality by repeated doubling and a CLT argument, then solves a finite-dimensional concave optimization problem built from the inverse Mills ratio to produce adaptive half-space tilts that force all shifted thresholds to equal the target threshold c . That yields the orthant-probability comparison

$$\mathbb{P}_{N(0,R)}(\max_i X_i \leq c) \geq \Phi(c)^m,$$

which is exactly the stochastic domination needed for the main theorem. filecite turn0file0



Step-by-step mapping of the proof to prior literature

The table below maps the manuscript's major steps to the closest identifiable prior art.

Manuscript step	Closest prior source	Match to prior literature	Assessment
Equal-energy weak simplex formulation for $m = n + 1$ codewords on the sphere, AWGN, ML decoding	Balitskiy–Karasev–Tsigler 2017 formulate the equal-energy simplex conjecture in exactly this geometric/ Gaussian-noise setting. Pastore 2023 gives the same weak-simplex problem statement and historical distinction from the strong conjecture. ⁹	Same problem statement, same surrounding history	Known
Proposition 3.1: decoding success $= \frac{e^{-\lambda^2/2}}{m} \mathbb{E} e^{\lambda M_G}$	Balitskiy–Karasev–Tsigler study the same functional $P(v_1, \dots, v_N) = \int \max_i e^{2x \cdot v_i - v_i ^2} dx$, which is the same ML/Gaussian-success functional up to normalization and scaling conventions. ¹⁰	Essentially the same integral identity in a different normalization	Known

Manuscript step	Closest prior source	Match to prior literature	Assessment
Simplex-to-independence normalization $R = \alpha G + J/m$ and the fact that the regular simplex gives $R = I_m$	I did not find this exact normalization packaged in this way in the searched literature. But it is a short linear-algebraic observation built on the standard regular-simplex Gram matrix and the standard Gaussian reformulation of simplex extremal problems. ⁶	Related ingredients are standard; this packaging may be original	Possibly novel, but elementary
Reduction to an orthant/CDF comparison for $N(0, R)$ versus i.i.d. Gaussians	Kabluchko–Litvak–Zaporozhets reduce only to an expectation comparison for Gaussian maxima, not to full stochastic domination. I did not find a prior exact match to the manuscript’s all- c orthant comparison in this simplex setting. ⁶	Stronger than the classical mean-width reformulation	Potentially novel
Lemma 5.1: symmetric Gaussian rectangle inequality	This is classical Šidák–Khatrı territory. Balitskiy–Karasev–Tsigler explicitly invoke Šidák for related Gaussian plank intersections, and recent work even studies refinements of the Šidák–Khatrı inequality. ¹¹	Same inequality class; manuscript gives an independent proof	Known
Corollary 5.2: bounded even log-concave factors are positively correlated	This is a standard layer-cake corollary of the rectangle inequality once superlevel sets are symmetric intervals. The manuscript’s proof is routine from Lemma 5.1. filecite turn0file0	Standard consequence rather than a new principle	Known / straightforward

Manuscript step	Closest prior source	Match to prior literature	Assessment
Theorem 6.1: centered one-dimensional log-concave factors with zero Gaussian barycenter satisfy a product inequality under arbitrary Gaussian correlation	Here the closest overlap is with the 2025–2026 centered-Gaussian literature. Nakamura–Tsuji prove Gaussian correlation for centered convex sets in 2025, and Milman–Nakamura–Tsuji in 2026 develop a Gaussian forward-reverse Brascamp–Lieb inequality for centered log-concave functions, explicitly saying their proof uses self-convolutions and the central limit theorem and that centered log-concave data had already been recognized as the right setting in earlier Nakamura–Tsuji work. ¹²	Not an exact quoted theorem match from the public abstracts, but very close in scope, method, and conceptual setting	Probably not fully novel by 2026
Doubling operator + CLT proof of Section 6	Milman–Nakamura–Tsuji explicitly describe their centered log-concave FRBL proof as using “self-convolutions and the Central Limit Theorem,” which is strikingly close to the manuscript’s doubling-and-CLT mechanism. ¹³	Very close methodological overlap	Strong prior analogue
Inverse-Mills-ratio map $H(s) = s + \phi(s)/\Phi(s)$, concave primitive $\mathcal{F}$, and adaptive tilt	I did not find a prior source in the searched literature using this exact inverse-Mills variational package to solve the simplex orthant comparison. Related centered Gaussian saturation frameworks exist in the 2024–2026 work, but not with this exact finite-dimensional half-line tilt construction in the sources I found. ¹⁴	The exact construction appears distinctive	Most likely novel part
Cameron–Martin shift step for the tilted half-lines	Finite-dimensional Gaussian shift identities are standard. The manuscript’s use is classical. filecite turn0file0	Standard tool	Known

Manuscript step	Closest prior source	Match to prior literature	Assessment
Final implication to the weak simplex conjecture	Pastore 2023 already claims a full proof of the weak simplex conjecture via a completely different route based on a relaxation over codebooks and decision regions, Gaussian centroids, and symmetry of adjacent regions. ²	Same theorem, different proof architecture	Not first claimed proof

Assessment of novelty and overlap

The manuscript is **not the first proof claim** of the weak simplex conjecture. That priority belongs, at least at the level of publicly posted proof claims, to Pastore’s 2023 arXiv paper. Pastore states the weak simplex conjecture, distinguishes it from the strong simplex conjecture, and claims a full proof by jointly optimizing the codebook and decision regions and then forcing a regular-simplex equilibrium through Gaussian-centroid symmetry arguments. ¹⁵

Nor is the manuscript the first route through related geometric reformulations. The classical expectation/max-width reformulation is already in Kabluchko–Litvak–Zaporozhets 2015, the $m = 4$ Gaussian-maximum case was solved in 2020, and Goldsmith posted a full mean-width proof claim in 2021 using spherical Voronoi partitions and centroid arguments. So the landscape already contains both low-dimensional solved cases and at least one full mean-width proof claim. ¹⁶

That said, the attached manuscript does seem **substantively new as a proof strategy**. Its central combination of:

- a normalization that sends the regular simplex to independence,
- a Section 6 product inequality for centered one-dimensional log-concave factors,
- and a Section 8–9 adaptive tilt driven by the inverse Mills ratio,

is not how Pastore, Sun–Hu–Lan, or Goldsmith proceed. On architecture alone, this is a different proof family. ¹⁷

The main qualification is Section 6. By July 2026, the centered Gaussian inequality literature has moved sharply in the direction the manuscript uses. Nakamura–Tsuji’s 2025 result extends Gaussian correlation to centered convex sets, and Milman–Nakamura–Tsuji’s 2026 paper develops a centered log-concave Gaussian forward-reverse Brascamp–Lieb theorem, explicitly highlighting centered log-concave inputs and a self-convolution/CLT proof strategy. That does not prove the manuscript’s exact theorem is literally already published in the same guise, but it significantly weakens any claim that the centered-log-concave correlation mechanism itself is wholly new. ¹²

So the cleanest novelty verdict is this:

As a theorem claim, the manuscript is not novel. As a proof architecture, it is partly novel. Its most plausibly original contribution is the inverse-Mills adaptive tilt plus the way it converts centered log-

concave positivity into the exact Gaussian orthant comparison needed for the simplex problem. Its least plausibly original contribution is the Section 6 centered product inequality, because by 2026 that territory overlaps strongly with the centered Gaussian FRBL literature. turn0file0

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Whether the manuscript reduces the conjecture to an already solved problem

The short answer is **no, not in the simple sense**.

It does **not** merely reduce the weak simplex conjecture to the simplex mean-width conjecture. In fact, Goldsmith explicitly remarks that the weak simplex conjecture implies the simplex mean-width conjecture, which confirms the direction of logical strength. The attached manuscript proves a full- λ classification statement, hence more than a first-moment assertion. 

It also does **not** appear to simply repackage the 2017 Balitskiy–Karasev–Tsigler reduction to minimizing the Gaussian measure of an outscribed simplex. Their paper reduces the weak simplex conjecture to a conjecture about Gaussian measure of simplices containing a fixed ball. The attached manuscript instead reduces the problem to a **functional inequality for centered log-concave one-dimensional factors**, and then solves the remaining threshold alignment by a new finite-dimensional variational argument. Those are different reductions. 

The closest thing to a reduction to a solved problem is more subtle: **Section 6 may now fall under the umbrella of already-solved centered Gaussian functional inequalities**. Milman–Nakamura–Tsuji’s 2026 theorem is broad enough that this is a real possibility, although I did not work out a one-line specialization from their theorem statement to the manuscript’s Theorem 6.1. So I would not say the manuscript “reduces the weak simplex conjecture to an already solved problem” in any immediate formal sense. I would say instead that **its main internal reduction lands in an area that, by 2025–2026, is already heavily covered by centered Gaussian correlation and FRBL machinery**. 

Gaps, dependencies, and the most important next checks

I did not find an obvious fatal mathematical error in a first-pass reading. The manuscript is unusually explicit about its dependency chain, and most delicate steps are actually proved rather than waved away. The singular-covariance passage in Section 6 is handled by approximation; the section on H , r , and $\mathcal{F}$ is computational but self-contained; and the Cameron–Martin shift is correctly phrased in finite dimensions without requiring invertibility. turn0file0

The first place I would scrutinize as a referee is **Theorem 6.1**. Not because it obviously fails, but because it is the heart of the argument and the place of strongest overlap with recent centered Gaussian inequality theory. A useful verification path would be:

1. try to derive Theorem 6.1 directly from the 2026 Gaussian forward-reverse Brascamp–Lieb theorem, or from the 2025 centered-convex-set Gaussian correlation result by approximation if possible;
2. if that succeeds, reclassify Section 6 as an application of existing theory rather than a new theorem;

3. if it does not, isolate exactly what is new in the manuscript's one-dimensional factor setting. 21

The second most important check is **Section 8's variational problem**. The manuscript proves existence and uniqueness cleanly using strict concavity and coercivity, and the stationarity equations are elementary. But this section is where the proof's originality is concentrated, so one should verify carefully that the optimizer's domain constraints and the identity

$$s + a - Ra = c1$$

really hold in the full generality needed and that no hidden regularity assumption is being used.

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The third check is **provenance rather than correctness**. The manuscript currently appears to omit many of the most relevant citations: Kabluchko–Litvak–Zaporozhets on Gaussian maxima/mean width, Balitskiy–Karasev–Tsigler on the weak simplex conjecture and Gaussian-simplex reductions, Sun–Hu–Lan on the $m = 4$ case, Goldsmith on all-dimensional mean-width claims, Pastore on the prior weak-simplex proof claim, and the 2025–2026 centered Gaussian inequality papers. If the manuscript is intended for serious submission, this omission materially weakens any novelty claim and should be fixed. 22

Prioritized sources and where each one matters

The sources below are the ones I would put first in any formal verification dossier.

Priority	Source	Why it matters	Link
Highest	Adriano Pastore, <i>A Proof of the Weak Simplex Conjecture</i> (2023)	Earlier full proof claim of the same theorem; essential for any novelty assessment	23
Highest	Alexey Balitskiy, Roman Karasev, Alexander Tsigler, <i>Optimality of codes with respect to error probability in Gaussian noise</i> (2017)	Modern survey of the weak simplex conjecture; contains the standard ML integral functional and the Gaussian-simplex reduction	10
Highest	Zakhar Kabluchko, Alexander Litvak, Dmitry Zaporozhets, <i>Mean width of regular polytopes and expected maxima of correlated Gaussian variables</i> (2015)	Canonical Gaussian maximum / mean-width reformulation; baseline comparison for what is already known	6
High	Wei Sun, Ze-Chun Hu, Guolie Lan, <i>The 4-D Gaussian Random Vector Maximum Conjecture and the 3-D Simplex Mean Width Conjecture</i> (2020)	Best concrete solved finite case of the Gaussian maximum formulation	24
High	Aaron Goldsmith, <i>A Proof of the Simplex Mean Width Conjecture</i> (2021)	Earlier all-dimensional proof claim for the weaker mean-width problem; also explicitly notes WSC $\Rightarrow$ SMWC	25

Priority	Source	Why it matters	Link
Highest	Shohei Nakamura, Hiroshi Tsuji, <i>The Gaussian correlation inequality for centered convex sets</i> (2025)	Key sign that “centered” Gaussian inequalities are already developed beyond the symmetric case	26
Highest	Emanuel Milman, Shohei Nakamura, Hiroshi Tsuji, <i>The Gaussian Conjugate Rogers-Shephard Inequality</i> (2026)	Most relevant recent overlap with Section 6: centered log-concave data, Gaussian FRBL, self-convolutions, CLT, Gaussian saturation	18
Useful background	Thomas Royen, <i>A simple proof of the Gaussian correlation conjecture extended to multivariate gamma distributions</i> (2014)	Landmark background on Gaussian correlation inequalities that underlie later centered developments	27

For formal publication-status and priority checking beyond public abstracts, the next databases to consult are **MathSciNet** and **zbMATH Open**, especially for reviews or citations of Pastore 2023, Goldsmith 2021, and the 2025–2026 Nakamura–Tsuji / Milman–Nakamura–Tsuji papers.

Overall verdict: the attached manuscript is **not the first proof claim** of the weak simplex conjecture, and several of its ingredients are already known or now sit very close to recent centered Gaussian inequality machinery. But it **does appear to offer a substantially different proof strategy**, and its most credible claim to originality is the **inverse-Mills adaptive-tilt route** that converts centered log-concave positivity into the exact stochastic comparison needed for the weak simplex conjecture. [filecite turn0file0](#) [28](#)

[1](#) [8](#) [15](#) [23](#) [28](#) A Proof of the Weak Simplex Conjecture

<https://arxiv.org/abs/2306.13478>

[2](#) [17](#) arxiv.org

<https://arxiv.org/pdf/2306.13478.pdf>

[3](#) [4](#) [9](#) [10](#) [11](#) arxiv.org

<https://arxiv.org/pdf/1701.07986.pdf>

[5](#) [13](#) [18](#) [20](#) [21](#) The Gaussian Conjugate Rogers-Shephard Inequality

<https://arxiv.org/pdf/2602.07981.pdf>

[6](#) [7](#) [16](#) [22](#) Mean width of regular polytopes and expected maxima of correlated Gaussian variables

<https://arxiv.org/abs/1511.08479>

[12](#) [26](#) The Gaussian correlation inequality for centered convex sets

<https://arxiv.org/abs/2504.04337>

[14](#) A generalized Legendre duality relation and Gaussian saturation

https://arxiv.org/abs/2409.13611?utm_source=chatgpt.com

[19](#) arxiv.org

<https://arxiv.org/pdf/2112.03393.pdf>

²⁴ [2008.04827] The 4-D Gaussian Random Vector Maximum Conjecture and the 3-D Simplex Mean Width Conjecture

<https://arxiv.org/abs/2008.04827>

²⁵ [2112.03393] A Proof of the Simplex Mean Width Conjecture

<https://arxiv.org/abs/2112.03393>

²⁷ A simple proof of the Gaussian correlation conjecture extended to multivariate gamma distributions

https://arxiv.org/abs/1408.1028?utm_source=chatgpt.com